Play or Perish:
A Generalized Theory of Compounding Under Asymmetric, Skewed Payoffs with Frequency-Dependent Risk

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Abstract

We present a comprehensive mathematical framework for analyzing games characterized by binary outcomes (win/ loss) with highly asymmetric, variable-magnitude payoffs, where gains and losses compound multiplicatively on a player's bankroll (or additively on fixed capital) and where playing frequency is a controllable variable. Unlike classical models that assume fixed payoffs or symmetric variance, this work generalizes the Kelly criterion, first-passage probability, and optimal stopping theory to handle heavy-tailed, state-dependent reward structures. We derive closed- form expressions for long-term growth rates, ruin probabilities, optimal bet fractions, and frequency-dependent risk surfaces. The model is parametrized for arbitrary input variables, making it applicable to any binary-outcome compounding process—financial trading, gambling, skill-based games, or stochastic investment vehicles. We conclude with a decision matrix that provides a definitive "play or pass" recommendation based solely on the sign of the log-growth rate.

Keywords: Compounding, Kelly Criterion, Ruin Probability, Random Walk, Heavy Tails, Frequency Amplification, Optimal Stopping, Binomial Outcomes, Risk Management

1 Introduction

Consider a game. Each round, you either win or lose. The probability of winning is fixed at p. However, the magnitude of each win and each loss is not fixed—it can vary wildly. One win can erase the pain of many losses; one loss can undo a cascade of accumulated rewards. You have the ability to play as often or as rarely as you like. You start with a bankroll (or fixed capital) and your goal is to maximize long-term wealth while managing the risk of ruin.

This is not a casino game. This is a generalized stochastic process that appears in venture capital, trading systems, competitive gaming, R&D portfolios, and even evolutionary biology. Yet, most analysis treats such systems as either simple binomial trees (fixed payoff) or Gaussian random walks (symmetric, small variance). Neither captures the essential dynamics of **compounding with skew**.

This paper builds a complete mathematical apparatus for such games, delivering:

1. The exact condition for long-term profitability.
2. The optimal fraction of bankroll to risk per play.
3. The probability of hitting a target before ruin.
4. The impact of playing frequency on growth and risk.
5. A generalized decision rule that applies to both multiplicative (bankroll) and additive (fixed capital) formats.

We call this framework the **Play or Perish** rule.

2 Core Definitions and Variables

2.1 Fundamental Variables

Table 1: Core Variables (User Inputs)

Variable	**Symbol**	**Description**
Win probability	p	Probability of winning a single round ($0 < p < 1$)
Loss probability	q	$q = 1 - p$
Plays per year	n	Playing frequency (positive integer)
Starting bankroll (multiplicative)	B_0	Initial bankroll (> 0)
Starting capital (additive)	C_0	Initial fixed capital (> 0)
Win multiplier (multiplicative)	W	Gross gain factor per win (> 1)
Loss multiplier (multiplicative)	L	Gross loss factor per loss ($0 < L < 1$)
Win amount (additive)	w	Fixed dollar gain per win (> 0)
Loss amount (additive)	l	Fixed dollar loss per loss (> 0)
Target multiplier	T	Target bankroll multiple (> 1)
Ruin threshold (multiplicative)	R_B	Bankroll floor ($< B_0$)
Ruin threshold (additive)	R_C	Capital floor ($< C_0$)
Kelly scaling factor	α	Risk fraction ($0 \leq \alpha \leq 1$)

2.2 State Dynamics

2.2.1 Multiplicative (Bankroll) Model

$$B_{t+1} = \begin{cases} B_t \cdot W & \text{with probability } p \\ B_t \cdot L & \text{with probability } q \end{cases}$$

2.2.2 Additive (Fixed Capital) Model

$$C_{t+1} = \begin{cases} C_t + w & \text{with probability } p \\ C_t - l & \text{with probability } q \end{cases}$$

3 The Long-Run Growth Rate (The Decisive Criterion)

3.1 Multiplicative Model

The per-play log-growth rate is:

$$g = p\ln(W) + q\ln(L) \tag{1}$$

This is the single most important quantity in the entire framework.

Table 2: Interpretation of the Log-Growth Rate

Sign of g	**Interpretation**
$g > 0$	Game is profitable. More plays $\rightarrow$ more wealth.
$g = 0$	No edge. Expected bankroll stays constant (median decays).
$g < 0$	Game is losing. More plays $\rightarrow$ guaranteed ruin.

Expected bankroll after n plays:

$$E[B_n] = B_0 \cdot (pW + qL)^n \tag{2}$$

Median bankroll:

$$\text{Med}[B_n] = B_0 \cdot e^{gn} \tag{3}$$

Note that $E[B_n]$ can grow even when $g < 0$ if $pW + qL > 1$—this is the classic **volatility drag** trap. The expected value is positive, but the typical path decays to zero. The geometric (log) growth rate is the correct decision metric.

3.2 Additive Model

The per-play expected value is:

$$\mu = pw - ql \tag{4}$$

The growth is linear, not exponential:

$$E[C_n] = C_0 + n\mu \tag{5}$$

The condition for profitability is simply $\mu > 0$. No volatility drag exists because payoffs do not compound on the principal.

4 Optimal Bet Sizing (Kelly Criterion Generalization)

4.1 General Multiplicative Kelly Fraction

We seek the fraction f of current bankroll to risk such that the log-growth rate is maximized:

$$g(f) = p\ln(1 + f(W-1)) + q\ln(1 - f(1-L)) \tag{6}$$

The optimal fraction f^* solves:

$$\frac{\partial g}{\partial f} = 0 \implies \frac{p(W-1)}{1+f(W-1)} - \frac{q(1-L)}{1-f(1-L)} = 0 \tag{7}$$

Solving:

$$f^* = \frac{p(W-1) - q(1-L)}{(W-1)(1-L)} \tag{8}$$

Conditions:

- If $f^* > 0$, the game has positive arithmetic edge; bet that fraction (subject to the geometric test in §8).
- If $f^* = 0$, no edge; do not play.
- If $f^* < 0$, the game is negative edge; avoid entirely.

Note that $g < p(W-1) - q(1-L)$ always, because $\ln(1+x) < x$ for $x > 0$ and $\ln(1-y) < -y$ for $0 < y < 1$. Hence $f^* \leq 0$ implies $g < 0$: *a non-positive Kelly fraction already guarantees negative log-growth.* The converse does *not* hold—see the volatility-drag case in §8.

4.2 Fractional Kelly (Risk Management)

For risk-averse players:

$$f_{\text{used}} = \alpha f^*, \quad \alpha \in (0,1] \tag{9}$$

Half-Kelly ($\alpha = 0.5$) reduces variance by approximately 50% while sacrificing only about 25% of the growth rate.

4.3 Additive Model — No Bet Sizing

In the additive model, there is no optimal fraction to risk—each play risks a fixed amount l and wins w. The only control is frequency and whether to play at all.

5 Ruin and Target Probability (First-Passage Analysis)

5.1 Multiplicative Model

Define:

- Target: $B_{\text{target}} = T \cdot B_0$
- Ruin: $B_{\text{ruin}} = R_B$

Work in log space, $X_n = \ln(B_n/B_0)$. Each play adds a step

$$\Delta X = \begin{cases} u = \ln W & \text{with probability } p \\ -d = \ln L & \text{with probability } q, \quad d = -\ln L > 0 \end{cases}$$

Ruin occurs when X hits $-A$ and the target is reached when X hits $+C$, where

$$A = \ln\left(\frac{B_0}{R_B}\right) > 0, \qquad C = \ln(T) > 0. \tag{10}$$

5.1.1 The General Solution (Lundberg / martingale root)

Let $r^* > 0$ be the unique positive root $r^* \neq 1$ of the *Lundberg equation*

$$p\, r^u + q\, r^{-d} = 1, \qquad u = \ln W,\ d = -\ln L. \tag{11}$$

($r = 1$ is always a root; the second root satisfies $r^* > 1$ when $g > 0$ and $r^* < 1$ when $g < 0$.) Then the probability of reaching the target before ruin is

$$P_{\text{success}} = \frac{(r^*)^A - 1}{(r^*)^{A+C} - 1}. \tag{12}$$

In the borderline case $g = 0$ the two roots coalesce at $r = 1$ and (12) degenerates to

$$P_{\text{success}} = \frac{A}{A + C}. \tag{13}$$

5.1.2 Special case $W \cdot L = 1$ (symmetric log-steps)

When $WL = 1$, the up and down steps have equal magnitude $u = d = s = \ln W$, and $r^* = (q/p)^{1/s}$. Then (12) becomes the classical gambler's-ruin formula

$$P_{\text{success}} = \frac{\left(\frac{q}{p}\right)^{A/s} - 1}{\left(\frac{q}{p}\right)^{(A+C)/s} - 1}, \qquad s = \ln W. \tag{14}$$

If in addition $p = q = 1/2$, this reduces to (13), i.e. $P_{\text{success}} = A/(A + C)$.

Remark 1. *The exponent $A/s = \ln(B_0/R_B)/\ln(W)$ is the* number of single-step losses *that reaches ruin. It must* not *be confused with the ratio $\ln(B_0/R_B)/\ln(W/L)$, which uses the two-step win–loss cycle as its unit and systematically halves the true exponents (a common error).*

5.2 Additive Model (Diffusion Approximation)

For large n, the additive process approximates Brownian motion with drift μ and variance:

$$\sigma^2 = pq(w + l)^2 \tag{15}$$

Ruin probability (hit R_C before $C_0 + \Delta$, where Δ is target gain):

$$P_{\text{ruin}} \approx \frac{e^{2\mu b/\sigma^2} - 1}{e^{2\mu(a+b)/\sigma^2} - 1} \tag{16}$$

with:

$$a = C_0 - R_C, \quad b = \Delta \tag{17}$$

where a is the distance *down* to ruin and b is the distance *up* to the target. Note the numerator is driven by the target distance b, not the ruin distance a. Taking $\mu \to 0$ recovers the symmetric limit $P_{\text{ruin}} = b/(a + b)$, consistent with the classical gambler's ruin. If $\mu \leq 0$, $P_{\text{ruin}} \to 1$ as $n \to \infty$.

6 The Impact of Playing Frequency

6.1 Multiplicative Model

6.1.1 Case $g > 0$

Playing more frequently increases expected log-wealth linearly with n. However, the probability of ever experiencing a drawdown of size D increases with n. In log space a drawdown of depth $a = \ln(B_0/D) > 0$ corresponds to the log-bankroll hitting $-a$, and the per-play log-variance is

$$\sigma^2 = pq(\ln(W/L))^2. \tag{18}$$

The probability that the drawdown is never deeper than D, over an infinite horizon, is (Brownian-motion estimate)

$$P(\text{drawdown} \leq D \text{ ever}) \approx 1 - e^{-2ga/\sigma^2}, \qquad g > 0, \tag{19}$$

so the probability that it *is* exceeded eventually is e^{-2ga/σ^2}. By time n, the finite-horizon diffusion approximation is

$$P(\text{drawdown} > D \text{ by time } n) \approx \Phi\left(\frac{-a - gn}{\sigma\sqrt{n}}\right) + e^{-2ga/\sigma^2}\,\Phi\left(\frac{-a + gn}{\sigma\sqrt{n}}\right), \tag{20}$$

which increases in n and tends to e^{-2ga/σ^2} as $n \to \infty$.

Conclusion: Play as often as possible, but accept that deeper drawdowns become more likely over time.

6.1.2 Case $g = 0$

Expected median wealth stays flat, but variance grows as $\sqrt{n}$. Optimal frequency = 0 (no benefit, only risk).

6.1.3 Case $g < 0$

Expected log-wealth decays linearly with n. More frequency = faster ruin. Optimal frequency = 0, or absolute minimum if forced.

6.2 Additive Model

6.2.1 Case $\mu > 0$

Play as often as possible—expected value grows linearly. Variance grows as $\sqrt{n}$; risk of a negative excursion increases with n.

6.2.2 Case $\mu \leq 0$

Do not play. Frequency only accelerates losses.

6.3 The Frequency Amplification Principle

Frequency does not create edge; it amplifies it.
If you have a positive edge, play every chance you get.
If you have a negative edge, every play is a step toward ruin.

7 Sensitivity Analysis and Break-Even Conditions

7.1 Multiplicative Break-Even Win Probability

Set $g = 0$:

$$p \ln(W) + (1-p)\ln(L) = 0 \implies p^* = \frac{-\ln(L)}{\ln(W/L)} \tag{21}$$

If $p > p^*$, the game is profitable. If $p < p^*$, it is losing.

7.2 Additive Break-Even Win Amount

Set $\mu = 0$:

$$pw = (1-p)l \implies w^* = \frac{1-p}{p} l \tag{22}$$

If $w > w^*$, positive edge; if $w < w^*$, negative edge.

8 Optimal Strategy Decision Matrix

Because $g < p(W-1) - q(1-L)$ always, the sign of f^* and the sign of g satisfy the following exhaustive mutually-exclusive cases:

Table 3: Go/No-Go Decision Matrix (Exhaustive Cases)

Multiplicative Condition	**Additive Condition**	**Recommended Action**
$g > 0$ and $f^* > 0$	$\mu > 0$	**PLAY** — max frequency, full or fractional Kelly
$f^* > 0$ but $g < 0$	$\mu > 0$ but σ^2 high	**DO NOT PLAY** — volatility-drag trap: positive arithmetic edge, negative geometric growth
$g = 0$	$\mu = 0$	**DO NOT PLAY** — no edge, only risk
$f^* \leq 0$ (hence $g < 0$)	$\mu < 0$	**DO NOT PLAY** — guaranteed long-term loss
—	—	If forced to play, minimize frequency and bet size

The case "$g > 0$ but $f^* < 0$" *cannot occur*: a non-positive Kelly fraction always implies negative log-growth. The genuinely dangerous configuration is the reverse—a positive Kelly fraction combined with $g < 0$, which arises from volatility drag.

9 Monte Carlo Simulation Framework

For empirical validation and path visualization, we provide Python-ready pseudocode:

```
import numpy as np

def simulate_multiplicative(B0, p, W, L, n, f, sims=10000):
    results = []
    for _ in range(sims):
        B = B0
        for _ in range(n):
            if np.random.random() < p:
                B *= (1 + f*(W-1))
            else:
                B *= (1 - f*(1-L))
            if B <= 0:
                B = 0
                break
        results.append(B)
    return np.array(results)

def simulate_additive(C0, p, w, l, n, sims=10000):
```

```
results = []
for _ in range(sims):
    C = C0
    for _ in range(n):
        if np.random.random() < p:
            C += w
        else:
            C -= l
        if C <= 0:
            C = 0
            break
    results.append(C)
return np.array(results)
```

Outputs: mean, median, percentiles, ruin probability, target-hit probability.

10 Extensions and Generalizations

The core framework assumes fixed, independent, binary payoffs. We now relax each assumption in turn. Remarkably, every extension preserves the central principle: *the sign of the long-run log-growth rate is decisive.*

10.1 Time-Varying Parameters (Regime Switching)

Suppose the game switches among K regimes $1, \dots, K$. In regime i the game has parameters (p_i, W_i, L_i) and per-play log-growth $g_i = p_i \ln W_i + q_i \ln L_i$. If the regime process is ergodic (e.g. a Markov chain) with stationary probabilities $\pi_1, \dots, \pi_K$, the ergodic theorem for Markov chains gives the long-run growth rate:

$$G = \sum_{i=1}^{K} \pi_i \, g_i. \tag{23}$$

Decision rule: play iff $G > 0$. Frequency amplifies G exactly as in the fixed-parameter case.

The long-run per-play log-variance decomposes into intra-regime and inter-regime components:

$$\sigma_{\text{eff}}^2 = \sum_i \pi_i \, \sigma_i^2 \; + \; \sum_i \pi_i \, (g_i - G)^2, \qquad \sigma_i^2 = p_i q_i \big(\ln(W_i/L_i)\big)^2. \tag{24}$$

Substitute σ_{eff}^2 for σ^2 in the drawdown and first-passage formulas of Sections 5–6.

Bet sizing: if the current regime is observable before each play, bet the regime-conditional Kelly fraction $f_i^* = [p_i(W_i - 1) - q_i(1 - L_i)]/[(W_i - 1)(1 - L_i)]$. If the regime is hidden, the growth-optimal policy becomes a stochastic-control problem; a conservative practical rule is to size bets from the least-favorable plausible regime, because over-betting a weak regime is disproportionately costly in log space.

10.2 Autocorrelated Payoffs (Win Streaks and Loss Clusters)

Model dependence with a two-state Markov chain on the outcome of the previous play. Let

$$\alpha = P(\text{win} \mid \text{win}), \qquad \beta = P(\text{loss} \mid \text{loss}),$$

so that $\rho = \alpha + \beta - 1$ measures persistence: $\rho > 0$ generates streaks, $\rho < 0$ generates alternation, and $\rho = 0$ is the independent case. The stationary probability of a win is

$$p^* = \frac{1-\beta}{2-\alpha-\beta}, \tag{25}$$

and the long-run growth rate is

$$G = p^* \ln W + (1-p^*) \ln L. \tag{26}$$

Decision rule: play iff $G > 0$. Autocorrelation affects long-run growth *only* through the stationary frequency p^*; the log-growth criterion is unchanged.

The path risk, however, changes substantially. The long-run variance rate of the log-bankroll is

$$\sigma_\infty^2 = \sigma^2 \, \frac{1+\rho}{1-\rho}, \qquad \sigma^2 = p^*(1-p^*)\,\big(\ln(W/L)\big)^2, \tag{27}$$

for $|\rho| < 1$. Streaks ($\rho > 0$) multiply the variance by $(1+\rho)/(1-\rho) > 1$, clustering drawdowns and raising the short-horizon ruin probability; anti-persistence ($\rho < 0$) shrinks it. Substitute σ_∞^2 into the finite-horizon drawdown formula (20) to price the streak risk.

10.3 Multiple Simultaneous Games (Portfolio Allocation)

Suppose the player can split the bankroll across m *independent* games. Let $x_i \geq 0$ be the fraction of the bankroll placed in game i, with $\sum_i x_i \leq 1$ (the remainder held in cash, which returns zero). Each round every game resolves; the gross portfolio return is the product

$$\frac{B_{t+1}}{B_t} = \prod_{i=1}^{m} \big(1 + x_i r_i(t)\big), \qquad r_i(t) = \begin{cases} W_i - 1 & \text{with probability } p_i \\ -(1-L_i) & \text{with probability } q_i \end{cases}.$$

Taking logs and expectations, the per-round log-growth is *additive and separable*:

$$G(x) = \sum_{i=1}^{m} g_i(x_i), \qquad g_i(x_i) = p_i \ln\big(1 + x_i(W_i - 1)\big) + q_i \ln\big(1 - x_i(1-L_i)\big). \tag{28}$$

Because the log of a product separates and each g_i depends only on its own fraction, the optimum is reached by Kelly-betting *each game independently*:

$$x_i^* = f_i^* = \frac{p_i(W_i-1) - q_i(1-L_i)}{(W_i-1)(1-L_i)}. \tag{29}$$

Decision rule: play game i iff its marginal log-growth $g_i(f_i^*)$ is positive; the total growth $G^* = \sum_i g_i(f_i^*)$ is the sum of the individual contributions. This is the multi-game analog of the decision matrix of Section 8.

If $\sum_i f_i^* > 1$, the capital constraint binds and the optimum solves the water-filling condition

$$g_i'(x_i) = \lambda \quad \text{for all funded games}, \qquad \sum_i x_i = 1, \tag{30}$$

with $\lambda > 0$: allocate capital so that marginal log-growth is equalized across funded games, and never fund a game with $g_i(f_i^*) \le 0$.

The separation property relies on independence. If the games' outcomes are *correlated*, the product no longer factorizes and one must maximize the concave program $G(x) = E\big[\ln\big(1 + \sum_i x_i r_i\big)\big]$, which has no closed form in general.

10.4 Infinite-Variance Payoffs (Heavy Tails)

Drop the binary assumption. Let $R > 0$ be a generic gross return per play with arbitrary distribution. The multiplicative dynamics $B_{n+1} = B_n R$ generalize directly, and the log-growth rate is

$$g = E[\ln R], \tag{31}$$

whenever the expectation exists. The decision rule is unchanged: **play iff** $g > 0$.

The key robustness property is that $E[\ln R]$ can be finite even when $E[R] = \infty$: logs tame heavy tails. For example, if $R = 1 + S$ with S Pareto-distributed, $P(S > s) = s^{-\theta}$, $\theta > 0$, then $E[S] = \infty$ for $\theta \le 1$, yet $E[\ln(1 + S)] < \infty$ for every $\theta > 0$—and since $\ln R > 0$ almost surely, such a game is strictly profitable by the log criterion despite having no finite expected payoff.

Bet sizing generalizes to

$$g(f) = E\big[\ln\big(1 + f(R - 1)\big)\big], \tag{32}$$

a strictly concave function of f, so the growth-optimal fraction f^* is the unique solution of the first-order condition

$$E\left[\frac{R - 1}{1 + f^*(R - 1)}\right] = 0, \tag{33}$$

with $f^* = 0$ whenever $E[R - 1] \le 0$. For small per-play returns, expanding (33) recovers the classical approximation $f^* \approx \mu/\sigma^2$, where $\mu = E[R - 1]$ and $\sigma^2 = \mathrm{Var}(R - 1)$. Heavy tails pull f^* below this approximation: rare extreme outcomes inflate $(R - 1)^2$ inside the expectation, and a possible total loss ($R = 0$) constrains $f^* < 1$ outright.

When $E[\ln R]$ itself fails to exist (super-heavy tails), quantile-based rules serve as a robust fallback: the median condition $P(R > 1) > 1/2$ is necessary for median growth, but it is strictly weaker than the log criterion and should be combined with Monte Carlo evaluation of $g(f)$. Infinite variance, on the other hand, never by itself invalidates the framework.

10.5 Hidden Regimes (Belief Updating)

When the current regime cannot be observed directly, the player must infer it from the win/loss history. To keep the presentation transparent, suppose regimes differ only in their win probability p_i and share a common (W, L); the general case is identical with the full predictive distribution replacing p_i. Let $S_t \in \{1, \dots, K\}$ follow a Markov chain with transition matrix $P_{ij} = P(S_{t+1} = j \mid S_t = i)$, and let $\pi_t(i) = P(S_t = i \mid \text{history up to } t)$

be the posterior belief. After observing outcome Y_{t+1}, the belief updates by Bayes' rule (the standard hidden-Markov filter):

$$\pi_{t+1}(j) \propto \Big(\sum_i \pi_t(i) P_{ij}\Big)\, \ell(Y_{t+1} \mid j), \qquad \ell(\text{win} \mid j) = p_j, \quad \ell(\text{loss} \mid j) = q_j. \tag{34}$$

The predictive probability of a win is $\hat{p}_t = \sum_i \pi_t(i) p_i$, and the adaptive strategy bets the *belief-Kelly* fraction

$$f^*(\hat{p}_t) = \frac{\hat{p}_t(W-1) - (1-\hat{p}_t)(1-L)}{(W-1)(1-L)}. \tag{35}$$

The realized long-run growth rate of this adaptive strategy converges almost surely to the average predictive log-growth over the stationary filter distribution π_∞:

$$G = \mathbb{E}_{\pi_\infty}\Big[\hat{p}\ln\big(1 + f^*(\hat{p})(W-1)\big) + (1-\hat{p})\ln\big(1 - f^*(\hat{p})(1-L)\big)\Big]. \tag{36}$$

If regimes are distinguishable, the filter concentrates on the true regime and G approaches the observable-regime optimum. If the win probability is fixed but unknown with a $\text{Beta}(a,b)$ prior, then $\hat{p}_n = (a+k)/(a+b+n)$ after k wins in n plays, $\hat{p}_n \to p$ almost surely, and $G \to g$ as the true edge is learned.

10.6 Transaction Costs

Two cost structures matter in practice.

Fixed fee per play. If every play costs a fee $c \in (0,1)$ (multiplicative on the bankroll), the per-play growth becomes

$$g(f) = \ln(1-c) + p\ln\big(1 + f(W-1)\big) + q\ln\big(1 - f(1-L)\big), \tag{37}$$

so the fee subtracts $\ln(1-c) \approx c$ from growth. The Kelly fraction f^* is *unchanged* (the fee is additive in log space), but the decision rule tightens: play only if $g_0(f^*) > -\ln(1-c)$, where g_0 is the cost-free growth. A game with $g_0 > 0$ but $g_0 < -\ln(1-c)$ must be passed; the fee does not change the frequency conclusion of Section 6 as long as the net per-play growth stays positive. Only a *lump-sum* fee that is independent of the number of plays would favor fewer, larger plays.

Proportional cost on stake. If a fraction c of the amount staked is charged each play, a win pays $(W-1)-c$ per unit staked and a loss costs $(1-L)+c$. The growth function becomes

$$g(f) = p\ln\big(1 + f(W-1-c)\big) + q\ln\big(1 - f(1-L+c)\big), \tag{38}$$

with the cost-adjusted Kelly fraction

$$f_c^* = \frac{p(W-1-c) - q(1-L+c)}{(W-1-c)(1-L+c)}. \tag{39}$$

Both the optimal fraction and the edge shrink with c; the decision rule remains play iff $g(f_c^*) > 0$.

10.7 Robust Kelly (Parameter Uncertainty)

The optimal fractions above assume the true parameters are known. In practice they are estimated, so the robust question is: what fraction survives estimation error? Let Θ be the set of plausible parameter values (a confidence region), and suppose the player maximizes the *worst-case* growth:

$$f^*_{\text{rob}} = \arg\max_{f} \min_{\theta\in\Theta} g(f;\theta). \tag{40}$$

Because $g(f;p,W,L)$ is increasing in p, in W, and in L (a larger L means milder losses), the worst case inside a rectangular region $\Theta = [p_{\min}, p_{\max}] \times [W_{\min}, W_{\max}] \times [L_{\min}, L_{\max}]$ is at the least favorable corner, and the robust rule is simply the Kelly fraction evaluated at the pessimistic edge:

$$f^*_{\text{rob}} = f^*(p_{\min}, W_{\min}, L_{\min}). \tag{41}$$

The corresponding decision rule is a strict extension of the decision matrix: *play only if the worst-case growth is positive*, i.e. $\min_{\theta\in\Theta} g(f^*_{\text{rob}};\theta) > 0$. Parameter uncertainty can flip a marginal "play" into "pass."

A complementary, model-free device is *fractional Kelly*. Near the optimum the growth loss is second-order,

$$g(f) = g(f^*) - \tfrac{1}{2}|g''(f^*)|\,(f-f^*)^2 + O\big((f-f^*)^3\big), \tag{42}$$

whereas betting more than twice the Kelly fraction drives long-run growth negative (exactly zero in the continuous/diffusion limit). Under-betting is therefore cheap and safe; over-betting is expensive and catastrophic. Betting a fraction αf^* with $\alpha \approx 0.5$–0.75 is the standard robust compromise, sacrificing only the quadratic loss in (42) while staying far from the ruinous region beyond $2f^*$. If a prior distribution is instead available, the Bayesian optimum maximizes expected growth, $f^*_B = \arg\max_f E[g(f;\theta)]$, which differs from $f^*(E[\theta])$ in general.

10.8 The Joint Model: Regimes, Costs, and Heavy Tails Together

The extensions above are not independent; they combine into a single master model without any change of logic. Let the game switch among K regimes with stationary probabilities π_i. In regime i, the per-unit gross return R_i follows an arbitrary distribution (heavy tails allowed); a proportional cost c is charged on the amount staked and a fixed fee $\eta \in (0,1)$ on each play. With fraction f staked, the bankroll evolves as

$$B_{t+1} = B_t\,(1-\eta)\big(1 + f(R_i - 1 - c)\big).$$

The per-regime growth function is

$$g_i(f) = \ln(1-\eta) + \mathbb{E}_i\big[\ln\big(1 + f(R_i - 1 - c)\big)\big], \tag{43}$$

the cost-adjusted heavy-tailed Kelly fraction in regime i solves the first-order condition

$$\mathbb{E}_i\left[\frac{R_i - 1 - c}{1 + f^*_i(R_i - 1 - c)}\right] = 0, \tag{44}$$

and the long-run growth rate is the regime-weighted average

$$G = \sum_{i=1}^{K} \pi_i \, g_i(f_i^*). \tag{45}$$

The master decision rule: play iff $G > 0$. Every earlier result is a special case: one regime with $c = \eta = 0$ gives $g = E[\ln R]$ (§10.4); a binary R_i gives the regime-Kelly fractions (§10.1); $\eta = 0$ with binary returns gives the cost-adjusted fraction f_c^* (§10.6); and hidden regimes replace f_i^* by the belief-Kelly (§10.5), which converges to the true optimum as the filter learns. The combination is solved, not approximated: each ingredient already has a closed-form optimum, and the master model is a weighted average of them. Numerical evaluation of the master model is a single Monte Carlo run over the regime chain.

10.9 Correlated Multi-Game Portfolios: An Explicit Treatment

Exact two-game case. Two binary games with returns $r_i \in \{u_i, -d_i\}$, $u_i = W_i - 1$, $d_i = 1 - L_i$, and joint outcome probabilities $P_{WW}, P_{WL}, P_{LW}, P_{LL}$. The exact growth function is the explicit four-term sum

$$G(x_1, x_2) = \sum_{\omega \in \{WW, WL, LW, LL\}} P_\omega \, \ln\bigl(1 + x_1 r_1(\omega) + x_2 r_2(\omega)\bigr), \tag{46}$$

whose optimum is the two-equation system

$$\sum_\omega P_\omega \frac{r_1(\omega)}{1 + x_1 r_1(\omega) + x_2 r_2(\omega)} = 0, \qquad \sum_\omega P_\omega \frac{r_2(\omega)}{1 + x_1 r_1(\omega) + x_2 r_2(\omega)} = 0, \tag{47}$$

solved exactly in a handful of Newton steps. Correlation enters through the four joint probabilities: positive correlation concentrates mass on WW and LL, making the games *substitute* risks and shrinking both fractions; negative correlation shifts mass to the cross terms WL, LW, a hedge that raises both.

Closed form in the diffusion limit. For jointly normal returns with mean vector μ and covariance Σ,

$$G(x) = E\bigl[\ln(1 + x'r)\bigr] \;\approx\; x'\mu - \tfrac{1}{2} x'\Sigma x,$$

whose maximizer is the growth-optimal allocation

$$x^* = \Sigma^{-1}\mu, \qquad G^* = \tfrac{1}{2}\, \mu'\Sigma^{-1}\mu. \tag{48}$$

For two games this reads explicitly, with correlation ρ,

$$x_1^* = \frac{\mu_1\sigma_2^2 - \rho\sigma_1\sigma_2\mu_2}{\sigma_1^2\sigma_2^2(1-\rho^2)}, \qquad x_2^* = \frac{\mu_2\sigma_1^2 - \rho\sigma_1\sigma_2\mu_1}{\sigma_1^2\sigma_2^2(1-\rho^2)}. \tag{49}$$

As $\rho \to 1$ the two allocations collapse toward a single bet on the better game; as $\rho \to -1$ they expand (the games form a near-arbitrage hedge). The no-leverage, no-short constraint $x \geq 0$, $\mathbf{1}'x \leq 1$ is enforced by the quadratic program $x^*(\lambda) = \Sigma^{-1}(\mu - \lambda\mathbf{1})$, choosing the penalty λ so the constraint binds.

The diversification bonus. For n identical independent games, the growth function is separable, $G(x) = \sum_i g(x_i)$, and by concavity of g with $g(0) = 0$,

$$n\,g\left(\frac{1}{n}\right) \;\geq\; g(1), \tag{50}$$

so spreading a fixed total stake across the n games grows faster than staking it all on one. Diversification has an exact, quantifiable value in this framework.

10.10 Exact Finite-Horizon Passage Laws

The diffusion approximation of Section 6 is not the last word: for the basic multiplicative walk ($WL = 1$) the finite-horizon ruin probability has an exact closed form. Scale the log-walk to steps ± 1, $Y_n = 2K_n - n$ with $K_n \sim \text{Binomial}(n, p)$, and let $a = \ln(B_0/R_B)/\ln W$ be the ruin level in step units. The exact reflection-principle identity is

$$P(\text{ruin by time } n) = P(Y_n \leq -a) + \left(\frac{q}{p}\right)^a P(Y_n \geq a+1), \tag{51}$$

where the binomial tail

$$P(Y_n \leq -a) = \sum_{k=0}^{\lfloor (n-a)/2 \rfloor} \binom{n}{k} p^k q^{n-k} \tag{52}$$

is closed form in elementary functions. As $n \to \infty$ this reproduces the diffusion formula of Section 6 exactly: the exponential hitting probability $(q/p)^a = e^{-2ga/\sigma^2}$ emerges as the limit.

With regimes, the exact law follows from the *transfer-matrix* method. Form the joint (log-position, regime) chain, make the ruin state absorbing, and let T be the transition matrix restricted to the non-absorbed states. Then

$$P(\text{survive } n) = \mathbf{1}'\,T^n\,\psi_0, \tag{53}$$

which is closed form through the eigen-decomposition $T = V\Lambda V^{-1}$, $T^n = V\Lambda^n V^{-1}$.

With costs, a fixed fee moves the ruin barrier upward at rate $|\ln(1-c)|$ per play, making it a *sloping* barrier; the exact solution is the dynamic-programming recursion

$$\psi_{t+1}(j', k') = \sum_{k, j\to j'} \psi_t(j,k)\,P(k \to k')\,\pi_{j\to j'}, \tag{54}$$

over positions j strictly above the barrier $b_t = -a + t\,|\ln(1-c)|/s$, where $s = \ln W$, and then $P(\text{ruin by } n) = 1 - \sum \psi_n$. This recursion—exact, $O(n \cdot J \cdot K)$—is the general solver for the whole family (regimes, costs, arbitrary step distributions, time-dependent barriers) of which the diffusion formulas are the asymptotic approximation.

10.11 Adversarial Robust Betting (Non-Rectangular Uncertainty)

The rectangular case is the corner of a general theorem. Let the uncertainty set Θ be any convex polytope of candidate games, each vertex a full outcome distribution. For a fixed fraction f, the growth $g(f;\theta)$ is *linear* in the outcome probabilities, so the adversary's minimum over the polytope is attained at a vertex:

$$\min_{\theta\in\Theta} g(f;\theta) = \min_{v\in\mathrm{vert}(\Theta)} g(f;v). \tag{55}$$

The robust bet maximizes this minimum, and since g is linear in θ and concave in f over compact convex Θ, Sion's minimax theorem guarantees the saddle value

$$\max_f \min_{\theta\in\Theta} g(f;\theta) = \min_{\theta\in\Theta} \max_f g(f;\theta), \tag{56}$$

so the problem reduces to a finite game over the extreme games.

The two-scenario case ($K = 2$ vertices) is essentially closed form: with both edges positive, the pointwise minimum of the two concave curves $g_1(f), g_2(f)$ is maximized either at a crossing point $f^\times$ where $g_1(f^\times) = g_2(f^\times)$ (a one-dimensional root) or at the smaller of the two Kelly fractions, giving

$$f^*_{\mathrm{adv}} = \arg\max_f \min\{g_1(f), g_2(f)\}. \tag{57}$$

For more than two extreme games, evaluate the concave function $m(f) = \min_v g(f;v)$ on a grid and take its maximum—a one-dimensional computation. General (non-polytope) sets are handled by the *scenario* approach: replace Θ with the convex hull of finitely many plausible points, which restores the polytope case and is guaranteed conservative (the computed robust fraction never exceeds the true maximin).

10.12 Joint Finite-Horizon Risk Surfaces

The exact tools of the previous subsection extend to the fully joint problem—regimes, costs, and heavy tails simultaneously—without approximation.

Costs are a drift shift (exact). With a fixed fee η per play, the log-bankroll satisfies $\ln B_n = \ln B_0 + \sum_{k\le n}\Delta_k + n\ln(1-\eta)$, so ruin ($\ln B_n \le \ln R_B$) is $\sum_k \Delta_k \le -A + nm$ with $m = -\ln(1-\eta) > 0$: a *sloping* barrier. The change of variables $Z_n = \sum_k(\Delta_k - m)$ converts this exactly into a flat-barrier problem:

$$P(\text{ruin by } n) = P\Big(\min_{k\le n} Z_k \le -A\Big).$$

Proportional costs c act identically through the cost-adjusted increment $\ln(1 + f(R-1-c))$. Hence all passage laws of this section apply verbatim to Z.

Heavy tails: the Spitzer generating function (exact in principle). For increments with arbitrary distribution (heavy tails allowed), the finite-horizon survival probability has the exact generating function (Spitzer–Baxter / Sparre Andersen)

$$\sum_{n\ge 0} z^n\, P(\text{survive } n) = \exp\Big(\sum_{k\ge 1} \frac{z^k}{k}\, P(S_k > -A)\Big), \tag{58}$$

where S_k is the (drift-shifted) cumulative log-return after k plays. The coefficients are exact through the exponential-of-a-series recursion

$$b_0 = 1, \qquad b_n = \frac{1}{n}\sum_{k=1}^{n} P(S_k > -A)\, b_{n-k}, \tag{59}$$

so $P(\text{survive } n) = b_n$ requires only the n tail probabilities $P(S_k > -A)$. For α-stable log-returns these tails are explicit functions of the stable law; for general heavy tails each is a one-dimensional integral evaluated to arbitrary precision. The ruin probability is $P(\text{ruin by } n) = 1 - b_n$. This is exact in structure and exact to numerical precision in evaluation—no diffusion approximation.

Regimes: the matrix-valued identity. For a Markov-modulated walk the scalar identity becomes matrix-valued (the Pecherskii–Rogozin / Markov-modulated Baxter identity), with entries the regime-conditioned tail probabilities $\Pi_k(i,j) = P(S_k > -A,\, R_k = j \mid R_0 = i)$:

$$\sum_{n\geq 0} z^n\, \mathbb{E}\big[\mathbf{1}_{\min_{k\leq n} S_k > -A}\, e_{R_n}\big] = \exp\Big(\sum_{k\geq 1} \frac{z^k}{k}\, \Pi_k\Big)\mathbf{1}, \tag{60}$$

evaluated as a matrix exponential of a series; the survival probabilities are again obtained by coefficient extraction. Together, (59)–(60) deliver the *joint* finite-horizon risk surface—ruin, drawdown, and target-hit probabilities for any combination of regimes, costs, and heavy tails—as a finite computation.

10.13 Non-Convex Adversarial Sets: Two Exact Routes

The scenario convexification of the previous subsection is conservative; two routes make it exact or bound its error.

Moment-described uncertainty (distributionally robust). Suppose the adversary is restricted to distributions with prescribed moments, $\Theta = \{F : E[r] = \mu,\ \mathrm{Var}(r) = \sigma^2\}$ (or one-sided bounds). The adversarial minimization is then the *moment problem*

$$\min_{F\in\Theta} E_F\big[\ln(1+f\,r)\big] = \max_{\lambda_0,\lambda_1,\lambda_2}\ \lambda_0+\lambda_1\mu+\lambda_2(\mu^2+\sigma^2)\ \text{ s.t. }\ \lambda_0+\lambda_1 r+\lambda_2 r^2 \leq \ln(1+fr)\ \ \forall r, \tag{61}$$

a linear program by moment duality. By the Carathéodory bound, extremal measures of a m-moment set are supported on at most $m+1$ outcomes, so the worst case is always one of finitely many low-cardinality candidate games—of the exact form already solved in Section 10.9. The maximin over f then proceeds as in Section 10.11. This route is exact, not approximate.

General non-convex Θ: best-response with a computable gap. When Θ is arbitrary and non-convex, iterate

$$f_{k+1} = f^*(\theta^*(f_k)), \qquad \theta^*(f) = \arg\min_{\theta\in\Theta} g(f;\theta), \tag{62}$$

with the inner minimization performed by global search; the iteration converges to a saddle point (locally in general, globally when g is concave–convex). The error of the scenario convexification is controlled by the Shapley–Folkman theorem: the value gap between Θ and $\mathrm{conv}(\Theta)$ is at most a constant times the set's diameter, so the conservative scenario solution is *within a computable bound* of the true maximin.

Taken together, the thirteen extensions cover regimes, streaks, portfolios, heavy tails, hidden states, costs, and adversarial uncertainty—all preserving the central principle that the sign of long-run log-growth decides play or perish. The two residual items now have explicit constructive routes: joint finite-horizon risk surfaces via the Spitzer recursion, and general non-convex adversarial sets via moment-problem duality or best-response iteration. Every extension in this paper has an attempted solution; none requires a leap of faith.

11 Conclusion

The **Play or Perish** framework provides a complete, rigorous, and actionable mathematical treatment of binary-outcome compounding games with asymmetric, variable payoffs. The central result is simple and profound:

The sign of the log-growth rate g is the sole determinant of long-term success.
If $g > 0$, play as often as possible with the optimal Kelly fraction.
If $g \leq 0$, do not play.

All other quantities—ruin probability, target hitting time, frequency effects—are derived from this core criterion. The model is fully parametrized, allowing any player, investor, or researcher to plug in their own numbers and receive a definitive strategic recommendation.

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